

Math 526, S16

Quiz 4

Section:

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Given

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}.$$

Find A^{-1} using the Gaussian-Jordan Elimination.

$$\begin{aligned} [A|I] &= \begin{bmatrix} \boxed{1} & 1 & 2 & | & 1 & 0 & 0 \\ \textcircled{2} & 4 & 5 & | & 0 & 1 & 0 \\ 0 & 4 & 0 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & \boxed{2} & 1 & | & -2 & 1 & 0 \\ 0 & \textcircled{4} & 0 & | & 0 & 0 & 1 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 2 & \textcircled{1} & | & -2 & 1 & 0 \\ 0 & 0 & \boxed{-2} & | & 4 & -2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & \textcircled{2} & | & 1 & 0 & 0 \\ 0 & 2 & 0 & | & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \boxed{-2} & | & 4 & -2 & 1 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} \textcircled{1} & 0 & 1 & | & 5 & -2 & \frac{3}{4} \\ 0 & \boxed{2} & 0 & | & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -2 & | & 4 & -2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 5 & -2 & \frac{3}{4} \\ 0 & 2 & 0 & | & 0 & 0 & \frac{1}{2} \\ 0 & 0 & -2 & | & 4 & -2 & 1 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 5 & -2 & \frac{3}{4} \\ 0 & 1 & 0 & | & 0 & 0 & \frac{1}{4} \\ 0 & 0 & 1 & | & -2 & 1 & -\frac{1}{2} \end{bmatrix} = [I|A^{-1}] \\ &\Rightarrow A^{-1} = \begin{bmatrix} 5 & -2 & \frac{3}{4} \\ 0 & 0 & \frac{1}{4} \\ -2 & 1 & -\frac{1}{2} \end{bmatrix} \end{aligned}$$

Problem 2. (4 points) Let $A = \begin{bmatrix} 3 & 3 & 4 \\ 1 & -1 & 3 \\ -2 & 4 & -1 \end{bmatrix}$. Find the LU factorization of A.

$$A = \begin{bmatrix} \boxed{3} & 3 & 4 \\ \textcircled{1} & -1 & 3 \\ -2 & 4 & -1 \end{bmatrix} \xrightarrow{l_{21} = \frac{1}{3}} \begin{bmatrix} \boxed{3} & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ \textcircled{-2} & 4 & -1 \end{bmatrix}$$

$$\xrightarrow{l_{31} = -\frac{2}{3}} \begin{bmatrix} 3 & 3 & 4 \\ 0 & \boxed{-2} & \frac{5}{3} \\ 0 & \textcircled{6} & \frac{5}{3} \end{bmatrix} \xrightarrow{l_{32} = -3} \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ 0 & 0 & \frac{20}{3} \end{bmatrix} = U$$

$\Rightarrow A = LU$ where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ -\frac{2}{3} & -3 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 3 & 3 & 4 \\ 0 & -2 & \frac{5}{3} \\ 0 & 0 & \frac{20}{3} \end{bmatrix}$$